
MACHINE LEARNING-BASED ANTIBODY HUMANIZATION PREDICTION AND FRAMEWORK REGION ANALYSIS

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ABSTRACT

Antibody humanization is an important step in therapeutic antibody development to reduce immune responses caused by non-human antibodies. In this project, a machine learning-based platform was developed to predict antibody humanization using approximately 29,000 antibody sequences. A Random Forest classifier trained on amino acid composition features achieved an accuracy of 98.25%. Further analysis using IMGT-numbered antibody regions showed that framework regions contained most of the humanization signal, while CDR regions contributed less. FR1 was identified as the most informative framework region, and several hotspot residues associated with species-specific patterns were discovered. These findings were integrated into a mutation recommendation engine capable of suggesting potential humanization mutations. The platform demonstrates the application of machine learning and immunoinformatics in antibody engineering.

1 INTRODUCTION

Antibody humanization is a critical step in therapeutic antibody development, aimed at reducing the immunogenicity of non-human antibodies while preserving their function. Traditional humanization approaches are often time-consuming and rely heavily on expert knowledge.

Applying the theoretical concepts studied from the BEC-210 Immunology Course to architect this project, it explores the use of machine learning to identify sequence patterns associated with antibody humanization. Using a dataset of approximately 29,000 antibody sequences, a predictive model was developed to classify human-like antibodies and investigate the contribution of different antibody regions. Beyond prediction, the study identifies key framework residues associated with humanization.

2 METHODOLOGY

The dataset included 13,240 human, 5,896 camel, 3,484 rat, and 6,420 additional antibody sequences from other sources. After preprocessing and validation, the sequences were used for feature extraction and machine learning analysis.

The trained model was evaluated using classification accuracy and achieved strong predictive performance, a Humanization score was calculated to provide a statistical evidence about adaptability of the antibody of an unknown species. To understand the biological basis of the predictions, antibody sequences were numbered using the IMGT scheme and divided into Complementarity Determining Regions (CDRs) and Framework Regions (FRs).

Separate machine learning models were trained on CDR and framework sequences to compare their contributions to humanization. Further analysis was performed on individual framework regions (FR1–FR4) to identify the most informative region. Feature importance analysis was then used to identify key residue positions associated with human-like sequence patterns.

	sequence	label	species	humanization_score	cdr1	cdr2
0	GSLRLSCAASGFTFGDYAMHWVRQAPGKGLEWVSLISWDGDSTYYA...	1	human	0.985	GFTFGDYA	ISWDGDST TKDRGIFG
1	ASVKVSCASGYTFTSYGISWVRQAPGQGLEWMGWISAYNGNTNYA...	1	human	1.000	GYTFTSYG	ISAYNGNT ARGYGF
2	GSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTYYA...	1	human	1.000	GFTFSSYA	ISGSGGST AKARLKG
3	HVQLVESGGGLVQPGGSLRLSCAASGFAFSTHDSWVRQAPGKGLE...	0	camel	0.025	GFAFSTHD	IYSDGSDT VI
4	EVQLVESGGGLVQPGGSLKLSCAASGFTFSSYTMWVRQTPEKRL...	0	mouse	0.000	GFTFSSYT	ISNGGGST

Figure 1: The first five rows of dataframe.

3 RESULTS AND DISCUSSION

The Random Forest classifier achieved an accuracy of 98.25%, demonstrating that antibody humanization can be effectively predicted using sequence-derived features. To understand the source of this predictive power, separate models were trained on CDR and Framework regions.

Framework regions achieved an accuracy of 96.50%, while CDR regions achieved 79.50%, indicating that the majority of the humanization signal is contained within the framework regions. Further analysis of individual framework regions revealed that FR1 was the most informative region, achieving 96.75% accuracy independently.

Feature importance analysis identified several key FR1 positions (12, 14, 19, 20, and 21) that exhibited strong species-specific residue preferences. The results demonstrate that machine learning can not only predict antibody humanization but also identify biologically meaningful sequence patterns that may assist antibody engineering efforts.

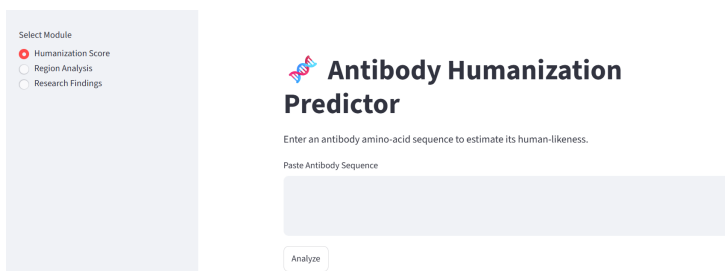


Figure 2: Antibody Analyzer Tool over Streamlit.

4 Conclusions

This project presents a machine learning-based platform for antibody humanization prediction and analysis. A Random Forest classifier achieved an accuracy of 98.25 in distinguishing human-like and non-human antibodies. Regional analysis demonstrated that framework regions contribute more strongly to humanization than CDR regions, with FR1 identified as the most informative framework region.

4.1 Analysis of Anomalies

While the Framework Regions (FRs) act as the conserved structural scaffold that supports and positions the hyper-variable Complementarity-Determining Regions (CDRs), the CDR directly bind to the target antigen. So from an immunological perspective, Complementarity-Determining Regions (CDRs) may be more accurately conceptualized as humanization-driving regions.

The primary explanatory factor for this divergence, is probably due to the Observed Antibody Space (OAS) dataset used, which containing corrupted sequences of non-human species like mice, which sequentially led to further overfitting in while training model and later analyzing framework. Overall, the primary motive of the study was to demonstrates how machine learning and immunology can be combined to support therapeutic antibody engineering and sequence optimization.